MADHUMITA SUBBIAH

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EDUCATION

Georgia Institute of Technology — **December 2027**
B.S. in Computer Science — *Atlanta, GA*

- Concentrations: Intelligence & Information Networks
- Relevant Coursework: Data Structures & Algorithms, OOP, Computer Architecture, AI, ML, Database Systems

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks/Libraries: React, Next.js, Node.js, Django, TensorFlow, PyTorch, Pandas, NumPy, scikit-learn
Tools/Platforms: AWS, PostgreSQL, MySQL, MongoDB, Docker, Git, Jira, REST APIs, CI/CD

EXPERIENCE

Raytheon Technologies | Software Engineering Intern — *Melbourne, FL* — **May 2026 – Present**

- AESA Weather Radar Systems
  - Improved turbulence detection range from 40 to 80 nautical miles by optimizing airborne radar detection algorithms.
  - Enhanced convective cell reflectivity analysis and turbulent weather detection through signal processing improvements.
- Internal AI Workflow Automation
  - Developed an agentic RAG-based LLM to automate Open Problem Report (OPR) generation for internal workflows.
  - Built retrieval and prompt pipelines using proprietary engineering data, reducing review time by 1–2 hours per task.

Saama Technologies | AI/ML Software Engineering Intern — *Campbell, CA* — **June – August 2025**

- Engineered Site360, a full-stack AI platform for clinical trial site selection, used to evaluate global rescue study feasibility.
- Engineered AWS S3-based ETL pipelines to process 2M+ patient records and integrate 20+ features for ML models.
- Built predictive site-ranking models using historical patient data, improving candidate site selection precision by 40%.

Health Informatics on FHIR Lab | Development Team Researcher — *Atlanta, GA* — **January 2025 – May 2026**

- Built Python ETL scripts to ingest UK-IPOP CSVs and map them into FHIR resources using Docker and HAPI FHIR tooling.
- Standardized 20+ healthcare attributes into FHIR profiles, reducing processing time for new datasets by 60%.
- Deployed scalable FHIR servers with REST endpoints to support querying of 1M+ anonymized patient records.

HEALTH-Engine Lab | Student Researcher — *Gainesville, FL* — **June – August 2024**

- Developed neural network models for radiotherapy dose optimization using voxel-based patient simulation data.
- Reduced model MSE from 5.29 to 0.0004 across 160 training iterations using Monte Carlo-generated radiation datasets.
- Co-authored an IEEE research paper on ML approaches for radiotherapy treatment planning and dosage optimization.

CS 1100: Freshman Leap Seminar | Teaching Assistant — *Atlanta, GA* — **August 2025 – May 2026**

- Mentored 200+ first-year CS students through 1:1 advising on academic planning, internships, and career development.
- Provided detailed feedback on 75+ resumes and reflective assignments to improve technical communication and clarity.

PROJECTS

BuzzOrg

- Leading a team of 7 to build a MERN-stack centralized application portal for 70+ Georgia Tech organizations and clubs.
- Implemented authentication, reviewer dashboards, and bulk decision tools, reducing review time per batch by 45%.

Waypoint

- Built an AI-powered travel tracking platform using Next.js and PostgreSQL to generate and share animated travel histories.
- Implemented DBSCAN-based trip inference pipelines using Apple/Google photo metadata and geospatial clustering.

Chronos

- Developed a full-stack scheduling application that automatically creates Google Calendar events from uploaded docs.
- Integrated a custom-trained named entity recognition model to extract dates, titles, and scheduling information from files.